

LOSO Experiment A1

Full sensors (flex+ypr+accel, both hands), INCLUDING Double_Nothing class

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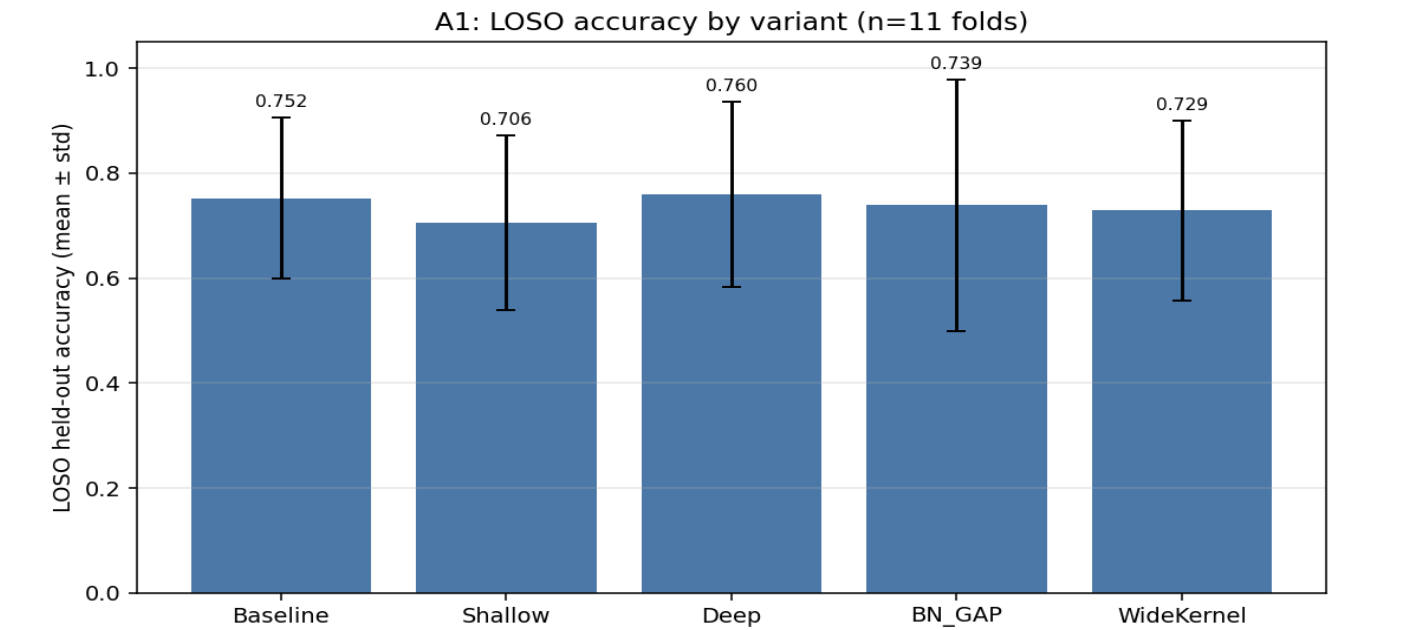
Experiment configuration

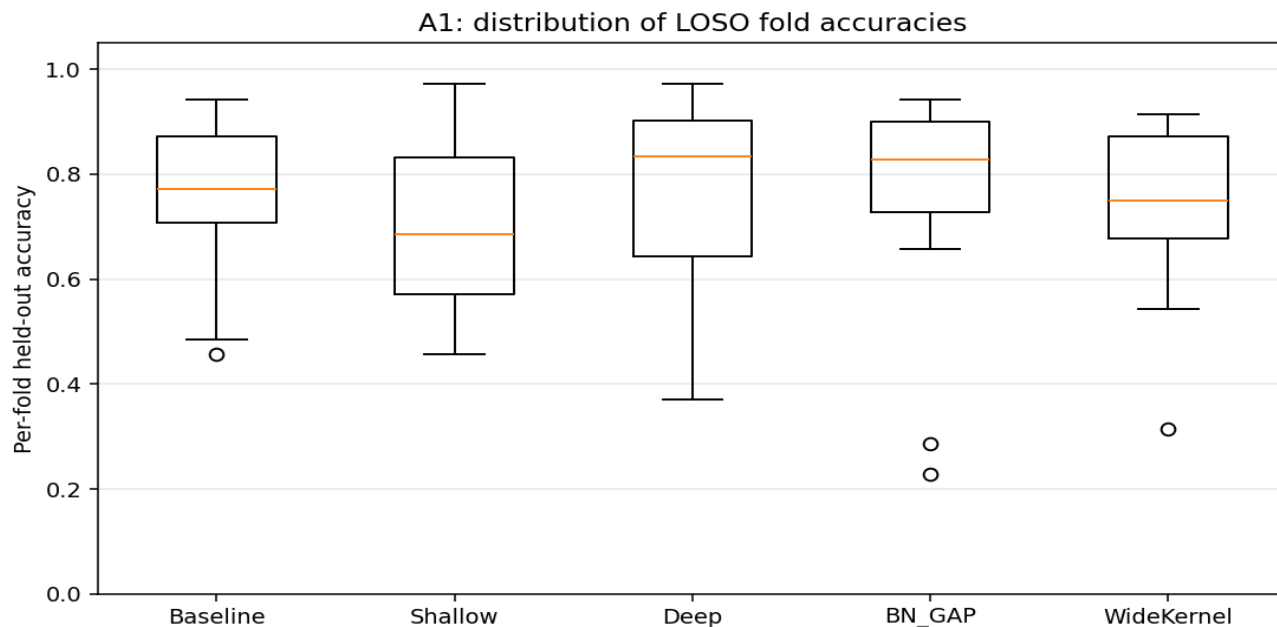
Class set	7 classes including Double_Nothing
Classes	Double_Lower, Double_Nothing, Double_Pistol_Recoil, Double_Raise, Double_Wiggle, Double_cmere, Drum_Roll
Sensor modalities	flex, YPR, accel
Hands	left, right
Sensor channels	176
Sequence length	90 steps (resampled)
Filter	Butterworth 10.0Hz order 4
Normalisation	minmax
Augmentation	amplitude_scale + baseline_offset, 3 copies
Inner CV folds	3
Epochs · batch size	8 · 16
LOSO subjects	11: 11_Mansh, 12_Marcus, 14_StephenV2, 15_Tash, 16_Bella, 17_Raquel, 1_Alان, 2_Alex, 3_Anghad, 4_Daniel, 6_Harry

LOSO summary across all subjects

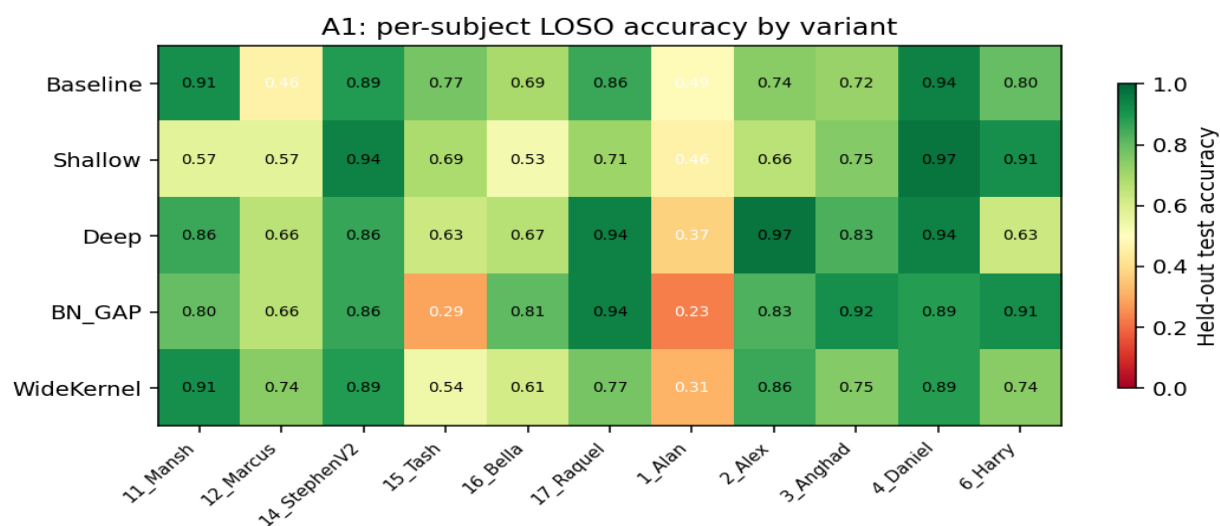
Variant	Params	LOSO mean	LOSO std	Min fold	Max fold	Inner CV mean (avg across folds)
Baseline	113,255	0.7525	0.1530	0.4571	0.9429	0.8677
Shallow	72,487	0.7059	0.1666	0.4571	0.9714	0.7436
Deep	204,007	0.7601	0.1762	0.3714	0.9714	0.8782
BN_GAP	57,319	0.7387	0.2392	0.2286	0.9429	0.9405
WideKernel	131,687	0.7292	0.1716	0.3143	0.9143	0.7763

Best variant by LOSO mean: **Deep**





Per-subject LOSO accuracy heatmap



Per-fold detail (held-out subject = each row)

Held-out subj	Train trials	Baseline	Shallow	Deep	BN_GAP	WideKernel
11_Mansh	353	0.914	0.571	0.857	0.800	0.914
12_Marcus	353	0.457	0.571	0.657	0.657	0.743
14_StephenV2	352	0.889	0.944	0.861	0.861	0.889
15_Tash	353	0.771	0.686	0.629	0.286	0.543
16_Bella	352	0.694	0.528	0.667	0.806	0.611
17_Raquel	353	0.857	0.714	0.943	0.943	0.771
1_Alan	353	0.486	0.457	0.371	0.229	0.314
2_Alex	353	0.743	0.657	0.971	0.829	0.857
3_Anghad	352	0.722	0.750	0.833	0.917	0.750
4_Daniel	353	0.943	0.971	0.943	0.886	0.886
6_Harry	353	0.800	0.914	0.629	0.914	0.743

Variant: Baseline

Params: 113,255

LOSO mean accuracy: 0.7525 ± 0.1530 (min 0.4571, max 0.9429)

Per-fold per-class held-out accuracy

Held-out	Double_Lower	Double_Nothing	Double_Pistol_Recoil	Double_Raise	Double_Wiggle	Double_cmere	Drum_Roll	Overall
11_Mansh	5/5	3/5	5/5	5/5	5/5	4/5	5/5	0.914
12_Marcus	5/5	0/5	1/5	5/5	0/5	5/5	0/5	0.457
14_StephenV2	4/5	3/6	5/5	5/5	5/5	5/5	5/5	0.889
15_Tash	5/5	3/5	5/5	5/5	4/5	0/5	5/5	0.771
16_Bella	5/5	0/6	5/5	5/5	0/5	5/5	5/5	0.694
17_Raquel	5/5	4/5	5/5	5/5	5/5	5/5	1/5	0.857
1_Alan	5/5	2/5	0/5	5/5	5/5	0/5	0/5	0.486
2_Alex	5/5	5/5	0/5	5/5	5/5	1/5	5/5	0.743
3_Anghad	2/5	3/5	5/5	5/5	0/5	6/6	5/5	0.722
4_Daniel	5/5	4/5	5/5	4/5	5/5	5/5	5/5	0.943
6_Harry	5/5	3/5	5/5	5/5	5/5	0/5	5/5	0.800
MEAN	0.927	0.536	0.745	0.982	0.709	0.636	0.745	0.752

Variant: Shallow

Params: 72,487

LOSO mean accuracy: 0.7059 ± 0.1666 (min 0.4571, max 0.9714)

Per-fold per-class held-out accuracy

Held-out	Double_Lower	Double_Nothing	Double_Pistol_Recoil	Double_Raise	Double_Wiggle	Double_cmere	Drum_Roll	Overall
11_Mansh	5/5	0/5	5/5	5/5	0/5	5/5	0/5	0.571
12_Marcus	4/5	0/5	4/5	5/5	2/5	5/5	0/5	0.571
14_StephenV2	5/5	4/6	5/5	5/5	5/5	5/5	5/5	0.944
15_Tash	5/5	0/5	5/5	5/5	4/5	1/5	4/5	0.686
16_Bella	4/5	0/6	5/5	0/5	0/5	5/5	5/5	0.528
17_Raquel	5/5	5/5	5/5	5/5	0/5	5/5	0/5	0.714
1_Alan	3/5	3/5	5/5	5/5	0/5	0/5	0/5	0.457
2_Alex	5/5	5/5	0/5	5/5	0/5	3/5	5/5	0.657
3_Anghad	5/5	1/5	5/5	5/5	0/5	6/6	5/5	0.750
4_Daniel	5/5	4/5	5/5	5/5	5/5	5/5	5/5	0.971
6_Harry	5/5	3/5	5/5	5/5	5/5	4/5	5/5	0.914
MEAN	0.927	0.442	0.891	0.909	0.382	0.782	0.618	0.706

Variant: Deep

Params: 204,007

LOSO mean accuracy: 0.7601 ± 0.1762 (min 0.3714, max 0.9714)

Per-fold per-class held-out accuracy

Held-out	Double_Lower	Double_Nothing	Double_Pistol_Recoil	Double_Raise	Double_Wiggle	Double_cmere	Drum_Roll	Overall
11_Mansh	5/5	0/5	5/5	5/5	5/5	5/5	5/5	0.857
12_Marcus	5/5	0/5	4/5	5/5	3/5	5/5	1/5	0.657
14_StephenV2	5/5	5/6	5/5	5/5	5/5	1/5	5/5	0.861
15_Tash	4/5	3/5	5/5	5/5	4/5	0/5	1/5	0.629
16_Bella	5/5	2/6	2/5	0/5	5/5	5/5	5/5	0.667
17_Raquel	5/5	5/5	5/5	5/5	3/5	5/5	5/5	0.943
1_Alan	4/5	4/5	0/5	5/5	0/5	0/5	0/5	0.371
2_Alex	5/5	5/5	4/5	5/5	5/5	5/5	5/5	0.971
3_Anghad	5/5	1/5	5/5	5/5	4/5	5/6	5/5	0.833
4_Daniel	5/5	3/5	5/5	5/5	5/5	5/5	5/5	0.943
6_Harry	5/5	2/5	5/5	5/5	0/5	4/5	1/5	0.629
MEAN	0.964	0.524	0.818	0.909	0.709	0.712	0.691	0.760

Variant: BN_GAP

Params: 57,319

LOSO mean accuracy: 0.7387 ± 0.2392 (min 0.2286, max 0.9429)

Per-fold per-class held-out accuracy

Held-out	Double_Lower	Double_Nothing	Double_Pistol_Recoil	Double_Raise	Double_Wiggle	Double_cmere	Drum_Roll	Overall
11_Mansh	5/5	0/5	3/5	5/5	5/5	5/5	5/5	0.800
12_Marcus	5/5	3/5	0/5	5/5	5/5	5/5	0/5	0.657
14_StephenV2	5/5	5/6	5/5	1/5	5/5	5/5	5/5	0.861
15_Tash	0/5	2/5	3/5	5/5	0/5	0/5	0/5	0.286
16_Bella	5/5	4/6	5/5	5/5	0/5	5/5	5/5	0.806
17_Raquel	5/5	5/5	5/5	5/5	5/5	5/5	3/5	0.943
1_Alan	0/5	5/5	0/5	0/5	3/5	0/5	0/5	0.229
2_Alex	5/5	5/5	0/5	5/5	5/5	5/5	4/5	0.829
3_Anghad	5/5	2/5	5/5	5/5	5/5	6/6	5/5	0.917
4_Daniel	5/5	1/5	5/5	5/5	5/5	5/5	5/5	0.886
6_Harry	5/5	3/5	5/5	5/5	5/5	4/5	5/5	0.914
MEAN	0.818	0.609	0.655	0.836	0.782	0.800	0.673	0.739

Variant: WideKernel

Params: 131,687

LOSO mean accuracy: 0.7292 ± 0.1716 (min 0.3143, max 0.9143)

Per-fold per-class held-out accuracy

Held-out	Double_Lower	Double_Nothing	Double_Pistol_Recoil	Double_Raise	Double_Wiggle	Double_cmere	Drum_Roll	Overall
11_Mansh	4/5	4/5	5/5	5/5	5/5	5/5	4/5	0.914
12_Marcus	5/5	0/5	5/5	5/5	2/5	5/5	4/5	0.743
14_StephenV2	5/5	3/6	5/5	5/5	4/5	5/5	5/5	0.889
15_Tash	0/5	3/5	5/5	5/5	5/5	0/5	1/5	0.543
16_Bella	5/5	2/6	2/5	5/5	0/5	5/5	3/5	0.611
17_Raquel	5/5	5/5	5/5	5/5	2/5	5/5	0/5	0.771
1_Alan	5/5	3/5	0/5	2/5	1/5	0/5	0/5	0.314
2_Alex	5/5	5/5	0/5	5/5	5/5	5/5	5/5	0.857
3_Anghad	5/5	2/5	5/5	5/5	0/5	6/6	4/5	0.750
4_Daniel	5/5	4/5	5/5	5/5	3/5	4/5	5/5	0.886
6_Harry	5/5	3/5	4/5	5/5	5/5	4/5	0/5	0.743
MEAN	0.891	0.603	0.745	0.945	0.582	0.782	0.564	0.729